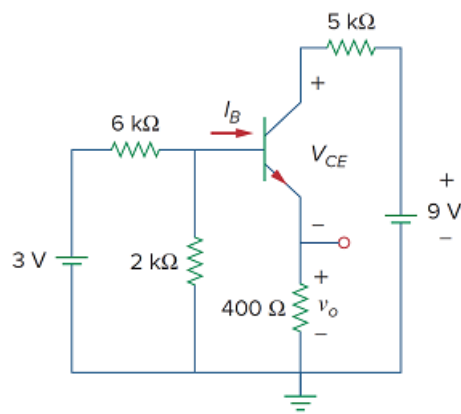


Problem

For the transistor circuit of Fig. 3.127, find I_B , V_{CE} , and v_o . Take $\beta = 150$, $V_{BE} = 0.7$ V.

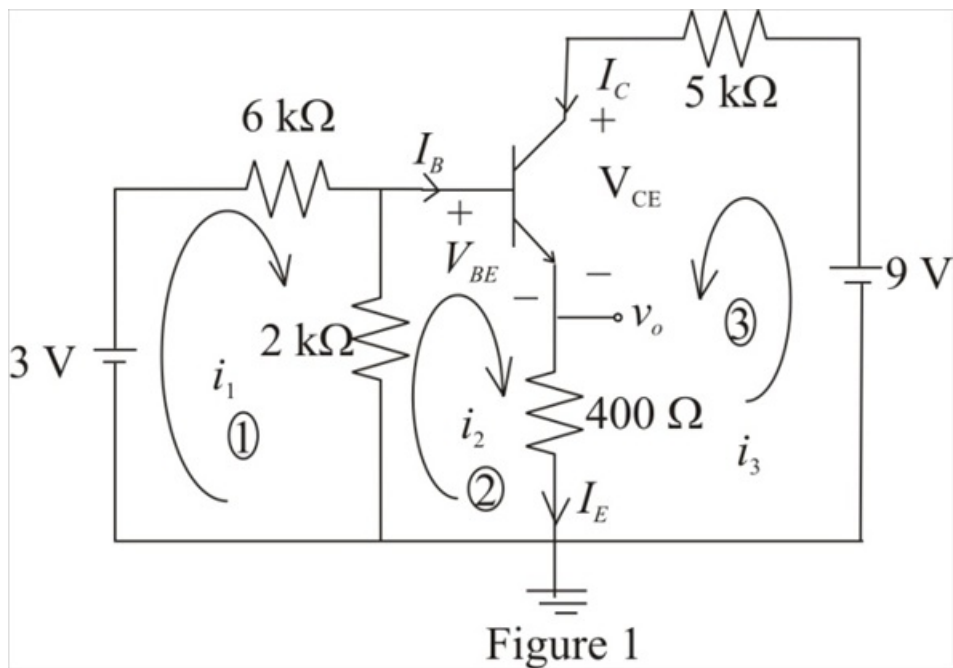
Figure 3.127:



Step-by-step solution

Step 1 of 7

Refer to Figure 3.127 in the text book.



Step 2 of 7

Notice that the current i_2 represents the base current I_B and current $i_2 + i_3$ represents the emitter current I_E .

Apply Kirchhoff's voltage law to loop-1 of the circuit shown in figure 1.

$$-3 + i_1(6 \text{ k}) + (i_1 - i_2)(2 \text{ k}) = 0$$

$$i_1(6 \text{ k} + 2 \text{ k}) - i_2(2 \text{ k}) = 3$$

$$i_1(8 \text{ k}) - i_2(2 \text{ k}) = 3$$

$$i_1(8 \text{ k}) = 3 + i_2(2 \text{ k})$$

$$i_1 = \frac{3 + i_2(2 \text{ k})}{8 \text{ k}}$$

Therefore, the expression for the current i_1 is, $\frac{3 + i_2(2 \text{ k})}{8 \text{ k}}$.

Step 3 of 7

Apply Kirchhoff's Voltage Law to loop-2 of the circuit shown in figure 1.

$$(i_2 - i_1)(2 \text{ k}) + V_{BE} + (i_2 + i_3)(400) = 0$$

Substitute $\frac{3 + i_2(2 \text{ k})}{8 \text{ k}}$ for i_1 , 0.7 V for V_{BE} and I_E for $i_2 + i_3$.

$$\left(i_2 - \frac{3 + i_2(2 \text{ k})}{8 \text{ k}} \right) (2 \text{ k}) + 0.7 + I_E(400) = 0$$

$$i_2(2 \text{ k}) - \frac{3 + i_2(2 \text{ k})}{4} + 0.7 + I_E(400) = 0$$

$$\frac{i_2(8 \text{ k}) - 3 - i_2(2 \text{ k}) + 2.8 + 1600I_E}{4} = 0$$

$$i_2(6 \text{ k}) - 0.2 + 1600I_E = 0$$

Substitute I_B for i_2 and $(1 + \beta)I_B$ for I_E .

$$I_B(6 \text{ k}) - 0.2 + 1600(1 + \beta)I_B = 0$$

Substitute 150 for β .

$$I_B(6 \text{ k}) - 0.2 + 1600(1 + 150)I_B = 0$$

$$I_B(6 \text{ k}) - 0.2 + 1600(151)I_B = 0$$

$$I_B(6 \text{ k}) - 0.2 + I_B(241.6 \text{ k}) = 0$$

$$I_B(247.6 \text{ k}) = 0.2$$

$$I_B = \frac{0.2}{247.6 \text{ k}} \\ = 0.8078 \mu\text{A}$$

Therefore, the value of the base current I_B is, $0.8078 \mu\text{A}$.

Step 4 of 7

Calculate the value of the emitter current I_E .

$$I_E = (1 + \beta)I_B$$

Substitute $0.8078 \mu\text{A}$ for I_B and 150 for β .

$$\begin{aligned} I_E &= (1 + 150)(0.8078 \mu) \\ &= (151)(0.8078 \mu) \\ &= 0.122 \text{ mA} \end{aligned}$$

Therefore, the value of the emitter current I_E is, 0.122 mA .

Step 5 of 7

Calculate the value of the collector current I_C .

$$I_C = \beta I_B$$

Substitute $0.8078 \mu\text{A}$ for I_B and 150 for β .

$$\begin{aligned} I_C &= (150)(0.8078 \mu) \\ &= 0.121 \text{ mA} \end{aligned}$$

Therefore, the value of the emitter current I_C is, 0.121 mA .

Step 6 of 7

Calculate the value of v_o .

$$v_o = (400)I_E$$

Substitute 0.122 mA for I_E .

$$\begin{aligned} v_o &= (400)(0.122 \text{ m}) \\ &= 48.8 \text{ mV} \end{aligned}$$

Therefore, the value of the voltage v_o is, 48.8 mV .

Step 7 of 7

Apply Kirchhoff's Voltage Law for the loop-3.

$$-9 + I_C(5 \text{ k}) + V_{CE} + v_o = 0$$

Substitute 0.121 mA for I_C and 48.8 mV for v_o .

$$-9 + (0.121 \text{ m})(5 \text{ k}) + V_{CE} + 48.8 \text{ m} = 0$$

$$-9 + 0.605 + V_{CE} + 48.8 \text{ m} = 0$$

$$-8.346 + V_{CE} = 0$$

$$V_{CE} = 8.346 \text{ V}$$

Therefore, the value of the voltage V_{CE} is, 8.346 V .